

# Kyeongseon (Ella) Kim

Ph.D. Student, School of Computing, KAIST

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## RESEARCH INTERESTS

How multimodal LLMs and agents encode and reason over *abstract visual representations* (vector graphics, GUIs, and symbols such as emoji and gestures) beyond natural images

## EDUCATION

### KAIST

Ph.D. Student, School of Computing (Advisor: Prof. Tae-Hyun Oh)

Daejeon, Korea

Sep. 2024 – Present

### Chung-Ang University

M.S. in Computer Science (Advisor: Prof. Joongheon Kim, now at Korea University)

Seoul, Korea

Sep. 2017 – Aug. 2019

B.S. in Computer Science and Engineering

Mar. 2013 – Feb. 2017

## PUBLICATIONS

“\*” denotes equal contribution

[C8] **IGG: A Benchmark for Interactive GUI Grounding under Visibility Constraints.**

**K. S. Kim\***, J. Son\*, and T.-H. Oh.

*Conference on Empirical Methods in Natural Language Processing (EMNLP 2026, Main)*

Also presented at ICML 2026 Workshop on Compositional Learning: Safety, Interpretability, and Agents

[C7] **SVHalluc: Benchmarking Speech–Vision Hallucination in Audio-Visual Large Language Models.**

C. Zhang, **K. S. Kim**, C. Liu, and T.-H. Oh.

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)*

[C6] **mEOL: Training-Free Instruction-Guided Multimodal Embedder for Vector Graphics and Image Retrieval.**

**K. S. Kim**, S.-E. Baek, J.-M. Lee, and T.-H. Oh.

*IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*

Accepted as a Round 1 paper (6.4% acceptance rate); Best Student Presentation Award, IPIU 2025

Also presented at ICCV 2025 Workshop on Graphic Design Understanding and Generation (GDUG)

[C5] **Zero-shot Depth Completion via Test-time Alignment with Affine-invariant Depth Prior.**

H. Lee, **K. S. Kim**, B.-K. Kwon, and T.-H. Oh.

*AAAI Conference on Artificial Intelligence (AAAI 2025)*

Qualcomm Innovation Fellowship 2025 winning paper

[J1] **Personalized Online Live Video Streaming Using Softmax-Based Multinomial Classification.**

**K. S. Kim**, D. Kwon, J. Kim, and A. Mohaisen.

*Vol. 9, No. 11, Article 2297 (Applied Sciences 2019)*

[C4] **Hardness on Style Transfer Deep Learning for Rococo Painting Masterpieces.**

**K. S. Kim**, D. Kim, and J. Kim.

*International Conference on Artificial Intelligence in Information and Communication (ICAII 2019)*

[C3] **Detecting Selfish Backoff Attack in IEEE 802.15.4 CSMA/CA Using Logistic Classification.**

J. Kim and **K. S. Kim**.

*International Conference on Ubiquitous and Future Networks (ICUFN 2018)*

[C2] **Recipient-oriented Transaction for Preventing Double Spending Attacks in Private Blockchain.**

H. Lee, M. Shin, **K. S. Kim**, Y. Kang, and J. Kim.

*IEEE International Conference on Sensing, Communication, and Networking (IEEE SECON 2018)*

[C1] **Self-Adaptive Machine Learning Operating Systems for Security Applications.**

**K. S. Kim**, D. Kwon, Y. Kim, J. Kim, and J. Kim.

*IEEE International Conference on Acoustics, Speech and Signal Processing (IEEE ICASSP 2018)*

[W1] **CuPS: Measuring Cultural Preference Signatures in LLM/VLM Agents and Their Steering by Profile Memories.**

**K. S. Kim**, G. Kim, J. Chang, H. Kim, and T.-H. Oh.

*Workshop on Culture × AI: Evaluating AI as a Cultural Technology (ICML 2026 Workshop)*

## COMMUNITY LEADERSHIP & EVENT ORGANIZATION

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|-------------------------------------------------------------------------------------------|----------------|
| <b>AGI Korea (formerly TensorFlow Korea)</b> – Co-organizer / Board Member (66K+ members) | 2020 – Present |
| <b>Google TensorFlow User Group (TFUG)</b> – Organizer (Global)                           | 2020 – Present |
| <b>TensorFlow Everywhere</b> – General Organizer (Global)                                 | Feb. 2021      |
| <b>Upstage × Naver Boostcamp AI Tech</b> – Organizer & Mentor                             | 2021 – 2022    |
| <b>Software Maestro</b> (Ministry of Science and ICT, Korea) – Mentor                     | 2022 – 2024    |
| <b>AI Weekly arXiv (Korea)</b> – Moderator                                                | 2022 – 2025    |
| <b>PR12 Paper Reading Group</b>                                                           | 2019 – Present |

## EXPERIENCE

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| <b>POSTECH</b>                                                                                                                                                                                                                                                                 | Pohang, Korea |
| <i>AI Research Scientist (Lab of Prof. Tae-Hyun Oh)</i>                                                                                                                                                                                                                        | 2023 – 2024   |
| <b>Upstage</b>                                                                                                                                                                                                                                                                 | Seoul, Korea  |
| <i>AI Contents &amp; Education Manager / AI Pack User Guide Project Manager</i>                                                                                                                                                                                                | 2022          |
| <b>LG Electronics (CTO Office AI Lab)</b>                                                                                                                                                                                                                                      | Seoul, Korea  |
| <i>AI Research Engineer (Team led by Prof. Joseph Lim, USC)</i>                                                                                                                                                                                                                | 2020 – 2022   |
| <ul style="list-style-type: none"><li>• Research topic: computer vision, scene graph generation</li><li>• Collaborated with LG Toronto AI Lab on floor plan generation for object placement</li><li>• Designed a driver action recognition system (Hyundai, Daimler)</li></ul> |               |
| <b>EASYWUM</b>                                                                                                                                                                                                                                                                 | Seoul, Korea  |
| <i>Tech Lead, Smart IoT Indoor Garden Startup</i>                                                                                                                                                                                                                              | 2016 – 2017   |

## AWARDS & SCHOLARSHIPS

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|------------------------------------------------------------------------------|----------------|
| <b>Korean Government Scholarship</b> (State-funded Ph.D. student), KAIST     | 2024 – Present |
| <b>Best Student Presentation Award</b> , IPIU (for the WACV 2026 paper [C6]) | 2025           |

## INVITED TALKS

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| <b>AI Talk for High School Sophomores and Juniors</b> , Jeonbuk Science High School  | 2023 |
| <b>MODUCON: AI Community Organizing</b> , Yonsei University / Lab for Everyone       | 2022 |
| <b>TensorFlow Everywhere Korea Keynote: What's New in TF 2.4</b> , Google            | 2021 |
| <b>Entrance Ceremony Speech</b> , Seoul High School                                  | 2021 |
| <b>AI Talk for High School Sophomores and Juniors</b> , Hanseong Science High School | 2021 |
| <b>1st KICS Lecture on TF-based Deep Learning Programming</b> , KICS                 | 2018 |
| <b>Deep Learning: Algorithms, Software, and Systems</b> , POSCO ICT                  | 2017 |
| <b>AI Fundamentals</b> , KT Education Center                                         | 2017 |

## TEACHING EXPERIENCE

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|----------------------------------------------------------------------------------------|------------|
| <b>Deep Learning Project Management Process</b> (Naver Boostcamp AI Tech) – Lecturer   | 2022       |
| <b>SW Health Contents Development</b> (College of Nursing, Chung-Ang Univ.) – Lecturer | 2018       |
| <b>Mobile Application Development</b> – TA & Lecturer                                  | 2017, 2018 |
| <b>Programming Language</b> – TA                                                       | 2017       |

## PATENTS

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| KR 10-1894314 (Granted): Method and Apparatus for Guiding a Recommended Parking Location ( <i>tech. transferred, Nov. 2024</i> ) |
| KR 10-2019-0005307: Method and System for Video Streaming Service                                                                |
| KR 10-2018-0089040: Recipient-oriented Transaction Verification Method and Apparatus for Preventing Double Spending              |
| KR 10-2018-0089039: Proof-of-Probability (PoP) based Blockchain Compensation Method and Apparatus                                |